



TECHNICAL PROFESSIONAL

**MARITIME - MARITIME ENGINEERING
AT SUPPORT LEVEL**

Grade 12

**Three-Term Budget Of Work (BOW)
for Learning Competencies**

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MARITIME
Grade 12**

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For Learning Competencies**

Please note that each TechPro elective is completed within one term if delivered at Grade 12.

Term Plan	
Elective	Maritime Engineering at Support Level
Performance Standard	The learners demonstrate the skills in the engine room piping system based on the Maritime industry standard. The learners demonstrate the skills in maintaining a safe engineering watch and safe working practices on the different shipboard scenarios. The learners perform shipboard maintenance and repair. The learners apply procedures in prevention of pollution, safe operation of equipment and machinery, fueling and oil transfer operations, bilge and ballast operations, and handling of stores.
Content Standard	- The learners demonstrate understanding of maintaining the safe engineering watch through monitoring and controlling the operation of a main propulsion and auxiliary machineries; the correct water level and steam pressure of the boiler; emergency procedures and emergency equipment in machinery spaces especially in the engine room operation including communication relevant to engine watchkeeping and the safe working practices and use of electrical equipment. - The learners demonstrate understanding of maintaining the correct water level and steam pressure of the boiler, emergency procedures in machinery spaces, operation of emergency equipment, safe engineering watch, communication relevant to engine watchkeeping, monitoring and controlling of an engine room watch, and understanding of the safe working practices and use of electrical equipment. - The learners demonstrate understanding of the shipboard maintenance and repair.

	The learners demonstrate understanding of anti-pollution equipment, operation of equipment and machinery, fueling and oil transfer operations, operating procedures of bilge and ballast system transfer operations, and handling of stores.
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Week	Learning Competency	Suggested Activities
1	- Discuss the shipping industry and its career opportunities. - Discuss the regulatory bodies and conventions in the shipping industry. - Discuss the stressors and emotional challenges of seafarers.	Shipping Industry Overview Report - Write a report highlighting the major sectors of the shipping industry and the types of careers available in each sector. Regulatory Research - Research major regulatory bodies and conventions (e.g., IMO, SOLAS) and summarize their roles in maritime operations. Convention Comparison Table - Create a table comparing the main international maritime conventions, their objectives, and the rules they enforce. Reflection Essay - Write a reflection on the common stressors seafarers face and strategies to cope with them. Stress Management Plan - Develop a plan with practical measures to manage stress and improve mental health

Week	Learning Competency	Suggested Activities
		for seafarers on long voyages.
2	- Discuss the shipboard organization and its functions according to the level of responsibility. - Discuss the type of ships and operations, parts, dimensions and terminologies used in ships direction. - Discuss the purpose charter party agreement in ship's trading services.	Organizational Chart - Create a chart showing the hierarchy of shipboard positions and their responsibilities. Role Analysis - Write a brief analysis explaining the duties of different levels of shipboard officers and crew members. Ship Type Classification - Prepare a chart classifying ships by type, function, and operation. Terminology Glossary - Compile a glossary of essential ship parts, dimensions, and navigation terms with definitions. Charter Party Summary - Write a summary explaining the purpose, types, and main clauses of charter party agreements. Case Study Analysis - Analyze a sample charter party agreement and identify key responsibilities of shipowners and charterers.
3	Discuss the safety and security code for the	Safety Code Report

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	protection of personnel, ship, cargo, and the environment. • Discuss shipboard machinery and equipment. • Perform gauge measurement. • Illustrate the different piping system in the engine room.	◦ Research the ISPS Code and write a report on its importance in protecting shipboard personnel, cargo, and the environment. • Scenario Safety Assessment ◦ Analyze a shipboard scenario and recommend safety and security measures according to the ISPS Code. • Machinery Identification Report ◦ Write a report identifying main and auxiliary shipboard machinery and their functions. • Equipment Diagram ◦ Draw a labeled diagram of shipboard machinery and equipment showing their placement and function. • Measurement Practice ◦ Perform gauge readings on simulated or real equipment and record the data accurately. • Data Analysis Exercise ◦ Analyze recorded gauge measurements, identify anomalies, and report findings. • Piping Diagram ◦ Draw diagrams illustrating the different piping systems in a ship's engine room. • System Function Table ◦ Create a table describing the purpose and

Week	Learning Competency	Suggested Activities
		function of each piping system in the engine room.
4	Use communication equipment relevant to engine room watchkeeping operations.	Equipment Operation Practice - Demonstrate the use of radios, alarms, and monitoring systems for engine room communication. Communication Scenario Simulation - Simulate a scenario where engine room communication is critical and document correct procedures.
5	Demonstrate emergency duties in response to emergency situations onboard as stated in the station bill.	Emergency Drill Practice - Practice the steps for responding to emergency scenarios onboard according to the station bill. Emergency Response Planning - Develop a plan detailing duties and responsibilities during emergencies on a ship.
6	Perform the operations of emergency equipment in the engine room in accordance with the standard safety procedures	Equipment Operation Practice - Operate engine room emergency equipment, such as fire pumps and alarms, following standard procedures. Emergency Equipment Checklist - Create a checklist for operating and

Week	Learning Competency	Suggested Activities
		maintaining emergency equipment in the engine room.
7	- Discuss watchkeeping duties for proper relief, maintenance, and handover of the watch - Perform monitoring, maintaining, and recording the safe operating parameters of the marine steam boiler. - Perform monitoring and controlling of an engine room watch.	Watchkeeping Procedure Notes - Write a detailed note on proper watchkeeping duties, including relief and handover procedures. Watchkeeping Scenario Simulation - Simulate a watch relief scenario and practice proper documentation, communication, and handover procedures. Boiler Monitoring Practice - Record and monitor marine steam boiler parameters under standard operating conditions. Parameter Analysis Exercise - Analyze boiler readings, detect anomalies, and suggest corrective actions. Engine Room Watch Simulation - Monitor engine room equipment, log readings, and maintain safe operating parameters. Watch Evaluation Exercise - Evaluate recorded data from the engine room watch and propose necessary adjustments.

Week	Learning Competency	Suggested Activities
8	- Perform safe working practices in shipboard in accordance with Safety Management System (SMS) manual. - Perform safe use of electrical equipment on shipboard maintenance and repair.	Safety Practices Reflection - Write a reflection on the importance of safe working practices in shipboard operations. SMS Compliance Exercise - Review a shipboard procedure and assess compliance with the SMS manual, identifying potential hazards. Electrical Safety Practice - Perform maintenance or repair on shipboard electrical equipment safely. Safety Procedure Documentation - Develop a step-by-step guide for safely maintaining and repairing electrical equipment on board.
9	Perform shipboard maintenance and repair of main and auxiliary machinery and equipment.	Activity 1: Machinery Maintenance Practice - Perform routine maintenance on main or auxiliary machinery following standard procedures. Maintenance Log Creation - Create a maintenance log documenting procedures, checks, and observed issues.
10 to 11	- Apply procedures in the prevention of pollution of the marine environment. - Perform safe operation of equipment and machinery in accordance with	Pollution Prevention Report - Write a report on procedures and best practices for preventing marine pollution onboard.

Week	Learning Competency	Suggested Activities
	manufacturers manual. • Perform fueling and oil transfer operations in accordance with standard safety procedures. • Perform bilge and ballast operations in accordance with standard safety procedures. • Perform handling of stores in accordance with standard safety procedures.	• Environmental Scenario Analysis ◦ Analyze a scenario where pollution could occur and recommend preventive measures. • Equipment Operation Practice ◦ Operate machinery following the manufacturer's manual to ensure safety and proper functioning. • Operation Procedure Checklist ◦ Create a checklist of steps to safely operate specific shipboard machinery. • Fuel Transfer Simulation ◦ Perform a simulated fueling or oil transfer operation while adhering to standard safety procedures. • Safety Procedure Review ◦ Develop a guide detailing safety precautions and procedures during fuel and oil transfer operations. • Bilge/Ballast Operation Practice ◦ Perform bilge and ballast operations safely, ensuring proper monitoring and recording. • Operation Analysis Exercise ◦ Analyze bilge and ballast procedures and identify potential hazards or inefficiencies. • Stores Handling Practice ◦ Safely handle, store, and organize

Week	Learning Competency	Suggested Activities
		shipboard supplies according to standard procedures. • Stores Safety Plan ◦ Develop a plan for safe handling, storage, and inventory of shipboard stores.
Suggested Performance Task	• Individual Activity: Piping System Diagram ◦ Task: Draw and label the engine room piping system, including main lines, valves, and connections, explaining their function according to maritime standards. • Group Activity: Piping System Simulation ◦ Task: In groups, assemble a model or mock-up of an engine room piping system, demonstrating proper flow, connections, and safety checks. • Individual Activity: Engineering Watch Reflection ◦ Task: Write a reflection on the responsibilities and procedures of maintaining a safe engineering watch, highlighting potential hazards and mitigation. • Group Activity: Watchkeeping Scenario Drill ◦ Task: In groups, simulate different shipboard scenarios requiring safe watchkeeping, including shift handover, emergency response, and routine monitoring. • Individual Activity: Engineering Watch Reflection ◦ Task: Write a reflection on the responsibilities and procedures of maintaining a safe engineering watch, highlighting potential hazards and mitigation. • Group Activity: Watchkeeping Scenario Drill ◦ Task: In groups, simulate different shipboard scenarios requiring safe watchkeeping, including shift handover, emergency response, and routine monitoring.	

Week	Learning Competency	Suggested Activities
		• Individual Activity: Maintenance Procedure Practice ○ Task: Perform a maintenance or repair task on a piece of shipboard equipment, following standard operating procedures and safety guidelines. • Group Activity: Maintenance Planning & Execution ○ Task: In groups, plan a shipboard maintenance schedule, perform assigned tasks on a model or simulation, and evaluate adherence to standards and safety. • Individual Activity: Safety & Pollution Prevention Report ○ Task: Write a report explaining procedures for pollution prevention, safe equipment operation, fueling, bilge, ballast, and handling of stores, highlighting safety measures. • Group Activity: Operational Simulation Exercise ○ Task: In groups, simulate a sequence of operations covering fueling, oil transfer, bilge and ballast operations, and handling of stores, ensuring compliance with safety standards and pollution prevention procedures.